MILITARY SPECIFICATION

INDICATOR, FREQUENCY CHANNEL ID-808(*)/ASQ

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide remote digital indication of the UHF communications preset channels when operated with Receiver-Transmitter, Radio RT-546/ASQ-19 or equivalent equipments.

1.2 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Control, Central C-3076/ASQ
Receiver-Transmitter, Radio RT-546/ASQ-19

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-L-25467	Lighting, Integral, Instrument
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-18309	Integrated Electronic Central AN/ASQ-19
MIL-D-4815	Detector, Leak, Mass Spectrometer Type MA-1

STANDARDS

Military

MS 33556	Case, Instrument, 3-1/4 inch Size, Standard Dimensions for
MS 33558	Numerals and Letters, Aircraft

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

FSC 5841

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3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and materials requirements shall be in accordance with Specification MIL-I-18309.

3.2 Design and Construction:

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-I-18309 for design, construction, and workmanship except in the paragraphs as modified below:

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 0.6 pound.

3.2.4.1 Operational Stability - The equipment shall operate with optimum performance for 2000 cycles, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 50,000 cycles with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.5.1 Cables and Connectors - The external connector shall be one Bendix type PTIH-14-19P, or approved equivalent, mounted on the rear of the indicator.

3.2.14 Warm-Up Time - The warm-up time for the equipment shall not exceed one (1) minute under standard conditions and two (2) minutes at extreme service conditions.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 *Channel Indication - The equipment shall provide indication of the UHF communications* preset channel (1 through 18), guard channel, and manual operation as selected by the Control, Central C-3076/ASQ-19.

3.4 Detail Requirements

3.4.1 Function - The equipment shall provide remote digital indication of the UHF communications preset channel.

3.4.2 Form Factor and Mounting - The equipment shall conform to the type of mounting and general form factor of MS 33556. The over-all dimensions of the equipment shall not exceed 1.645 by 1.33 by 5.906 inches.

3.4.3 Contents - The indicator shall contain the necessary automatic positioning device, digital indicator means, integral lighting and all other parts necessary to make the equipment function as a unit.

3.4.4 Controls, Internal - There shall be no internal, adjustable controls.

3.4.5 Integral Lighting - The equipment shall be integrally lighted in accordance with Specification MIL-L-25467.

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3.4.6 Power Shutter Mechanism - A shutter mechanism shall be provided for partially obscuring the indicator display. The shutter mechanism shall be excited and voltage excitation shall cause the shutter to disappear from view. The excitation voltage shall be obtained from the main power connection to the equipment.

3.4.6.1 Operating Current - The shutter mechanism operating current shall be 120 milliamperes maximum under standard conditions.

3.4.6.2 Lettering - The word OFF shall be on the shutter mask in block letters approximately 3/32 inch high on a red background. The letter style shall be in accordance with MS 33558.

3.4.7 Indications - The indicator shall display the following information in sequence in the order listed:

(1) UHF preset channel - indication 1 through 18.

(2) Manual frequency selection - indication M.

(3) Guard channel - indication G.

3.4.7.1 Numerals - All numerals and letters shall be in accordance with Military Standards MS 33558 (ASG) and approximately 1/4 inch high.

3.4.7.2 Visibility of Indications - With the shutter mechanism deflected the indications shall be legible from any point within a truncated polyhedran, the smaller end of which is the instrument aperture. The top, bottom and sides of the polyhedran are inclined at an angle of 45 degrees, 15 degrees, and 30 degrees respectively with a line perpendicular to the indicator mounting surface.

3.4.7.3 Sequence of Display - The indicator shall be marked such that the numerals in the window travel upward for increasing numerical values of channel. The equipment shall provide bidirectional operation. Each number or letter shall be driven by an intermittent motion type mechanism rather than a continuous type drive and shall be positively sequenced directly with the selected remote preset channel.

3.4.8 Duty Cycle - The maximum operating time of the equipment shall not exceed the following: two (2) seconds, on, four (4) seconds off.

3.4.9 Case - The indicator case shall be hermetically sealed. The case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures (soldering irons are considered to be standard tools). The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the component may be subjected. The case shall be designed so that it will meet the pressure requirements necessary when tested for leaks by means of a mass spectrometer type of helium leak detector in accordance with Specification MIL-D-4815.

3.4.9.1 Filling Medium - The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point - 65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately one atmosphere.

3.4.9.2 Leak Rate - The indicator shall be sealed such that the initial maximum detected leak rate at a pressure differential of approximately one atmosphere shall not exceed 1.0 micron cubic foot per hour.

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3.4.10 Power Requirements - The equipment shall give specified performance with the power sources listed in Specification MIL-I-18309 and below. The currents required shall not exceed the specified amounts under standard conditions.

(1) DC Power 28-Volt System

Current 0.3 amp

(2) DC Power 5-Volt System

Current 0.5 amp

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the test requirements of Specification MIL-I-18309 to determine compliance with all the applicable requirements except in the paragraphs as modified below:

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated for 100 operations without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Channel cycling	Random selection
Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's
Duty Cycle	Not to exceed that specified in Par. 3.4.8.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the 100 operations. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the operating test following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2.1 (3) Sampling Tests - Manufacturing reliability test as specified in 4.4.1.3 except that the test duration shall be 2000 operations

4.5 Life Test - Life test of the equipment shall consist of 50,000 operations with performance checks at approximately 10,000 cycle intervals.

5. PREPARATION FOR DELIVERY

5.1 General - Preparation for delivery shall be in accordance with Specification MIL-I-18309.

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6. NOTES

6.1 Use - This equipment is mainly intended to be used as part of the Integrated Electronic Central AN/ASQ-19. In this central it provides a visual, digital display of the channel to which the RT-546/ASQ-19 is tuned.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 volts dc and 5 ± 0.1 volts ac. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) Central Specification - Integrated Electronic Central AN/ASQ-19, Specification MIL-I-18309 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have preference over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of Specification MIL-I-18309. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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